

Producer Consumer Problem

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0, out = 0;

// Semaphores
sem_t empty, full, mutex;

// Producer Thread
void* producer(void* arg) {
    for(int i = 1; i <= 10; i++) {

        // Wait for empty slot
        sem_wait(&empty);

        // Enter critical section
        sem_wait(&mutex);

        // Produce item
        buffer[in] = i;
        printf("Produced: %d\n", i);
        in = (in + 1) % BUFFER_SIZE;

        // Exit critical section
        sem_post(&mutex);

        // Increase full count
        sem_post(&full);

        sleep(1);
    }
    return NULL;
}
```

```

int main() {
    pthread_t p, c;

    // 1. Initialize semaphores
    sem_init(&empty, 0, BUFFER_SIZE); // initially all slots empty
    sem_init(&full, 0, 0);             // initially no items
    sem_init(&mutex, 0, 1);            // binary semaphore

    // 2. Create threads
    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);

    // 3. Wait for threads to finish
    pthread_join(p, NULL);
    pthread_join(c, NULL);

    // Destroy semaphores
    sem_destroy(&empty);
    sem_destroy(&full);
    sem_destroy(&mutex);

    return 0;
}

```

```

Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
Produced: 6
Consumed: 6
Produced: 7
Consumed: 7
Produced: 8
Consumed: 8
Produced: 9
Consumed: 9
Produced: 10
Consumed: 10

```